CS372
FORMAL LANGUAGES & THE THEORY OF COMPUTATION

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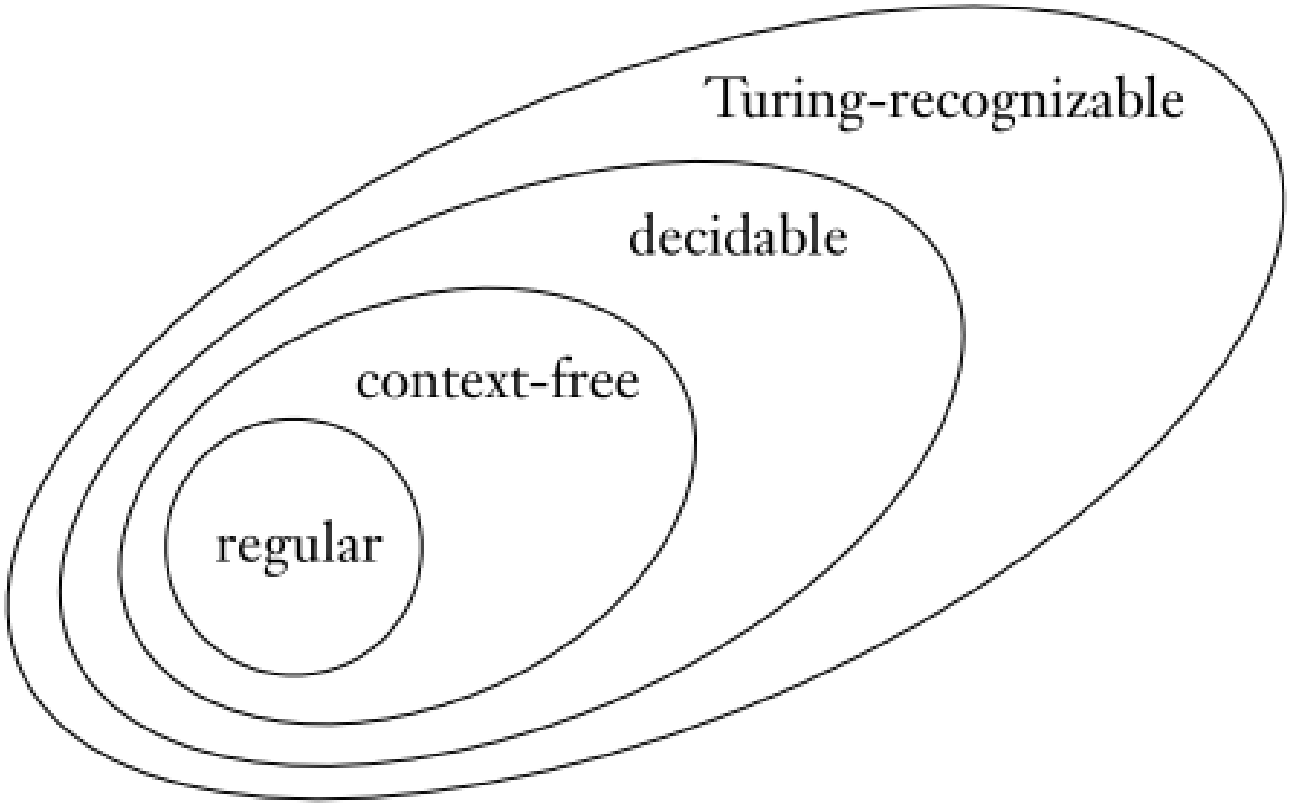
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Unit 9
Undecidability

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- Acceptance Problem for Turing machine
- The diagonalization method
- The halting problem

Relationship among classes of languages



Acceptance Problems for Turing Machines

Theorem 9.1

$A_{TM} = \{(M, w) \mid M \text{ is a TM and } M \text{ accepts } w\}$ is undecidable.

1. Note that A_{TM} is Turing-recognizable. This theorem when proved, shows that recognizers are more powerful than deciders.
2. We can encode TMs with strings just like we did for DFA's

Encoding aTuring Machine

- Encode Q using unary encoding:
  - For $Q = \{q_0, q_1, \ldots q_{n-1}\}$, encode q_i using $i + 1$ 0's, i.e., using the string 0^{i+1}. q_{accept} is encoded with 0^{n+1}, q_{reject} is encoded with 0^{n+2},
  - We assume that q_0 is always the start state.
- Encode Γ using unary encoding: $\Gamma = \{a_1, a_2, a_3\}$, where $a_1 = 0$, $a_2 = 1$, $a_3 = \square$. Encode a_1 using 0, a_2 using 00, a_3 using 000.
- Encode $\Delta = \{\Delta_1, \Delta_2\} = \{L, R\}$ using 0^t ($t = 1$ or $t = 2$)
- Encode δ. Each entry of δ, e.g., $\delta(q_i, a_j) = (q_k, a_m, \Delta_t)$ is encoded as

$$0^{i+1}10^{j+1}10^{k+1}10^{m+1}10^{t}$$

- Separate entries of δ with 11, the code for the whole machine begins with 111 and ends with 111 .

Example for TM Encoding

- TM for language L = $\{0^{2n} \mid n \geq 0\}$ with the following state diagram is

111

0101001010011

0100100001001011

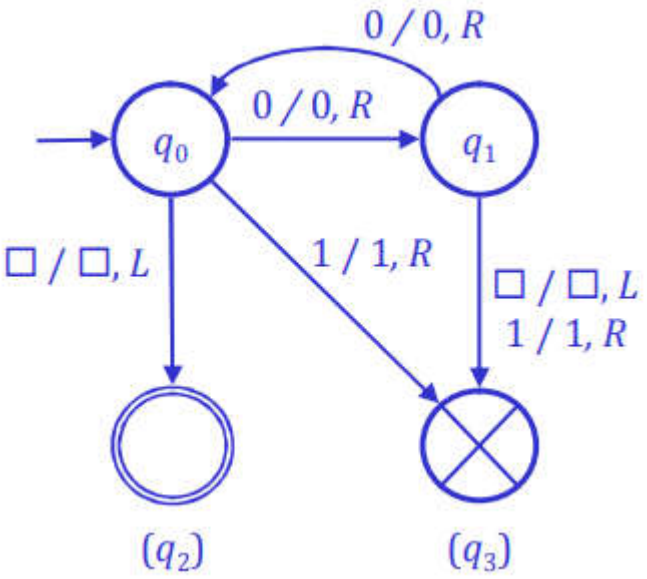
0100010001000101011

0010101010011

0010010000100100111

0010001000010010

111



Acceptance Problems for Turing Machines

- The TM U recognizes A_{TM}
- U = "On input (M, w) where M is a TM and w is a string:
  1. Simulate M on w
  2. If M ever enters its accepts state, *accept*; if M ever enters its reject state, *reject*.
- Note that if M loops on w, then U loops on (M, w), which is why it is NOT a decider!
- U can not detect that M halts on w .
- In some documents A_{TM} is also known as the Halting Problem
- U is known as the Universal Turing Machine because it can simulate every TM (including itself!)

The diagonalization method

- Let A and B be any two sets (not necessarily finite) and f be a function from A to B.
- f is one-to-one if $f(a) \neq f(b)$ whenever $a \neq b$.
- f is onto if for every $b \in B$ there is an $a \in A$ such that $f(a) = b$.
- We say A and B are the same size if there is a one-to-one and onto function $f : A \longrightarrow B$.
- Such a function is called a correspondence for pairing A and B.
  - Every element of A maps to a unique element of B
  - Each element of B has a unique element of A mapping to it.

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The diagonalization method

- Let N be the set of natural numbers $\{1, 2, \ldots\}$ and let E be the set of even numbers $\{2, 4, \ldots\}$.
- $f(n) = 2n$ is a correspondence between N and E.
- Hence, N and E have the same size (though $E \subset N$).
- A set A is countable if it is either finite or has the same size as N.
- $Q = \{ \frac{m}{n} \mid m, n \in N \}$ is countable!
- Z the set of integers is countable:

$$f(n) = \begin{cases} \frac{n}{2} & n \text{ even} \\ -\frac{n+1}{2} & n \text{ odd} \end{cases}$$

The diagonalization method

THEOREM

R *is uncountable*

PROOF.

Assume f exists and every number in R is listed.

Assume $x \in$ R is a real number such that x differs from the jth number in the jth decimal digit.

If x is listed at some position k , then it

differs from itself at k^{th} position; otherwise the premise does not hold

f does not exist

n	$f(n)$
1	3.14159. . .
2	55.77777. . .
3	0.12345. . .
4	0.50000. . .
.	
=	.4527 . . .

defined as such, can not be on this list.

Diagonalization over languages

Corollary

Some languages are not Turing-recognizable.

Proof

For any alphabet Σ, Σ^* is countable. Order strings in Σ^* by length and then alphanumerically, so $\Sigma^* = \{s_1, s_2, \ldots, s_i, \ldots\}$

The set of all TMs is a countable language.

Each TM M corresponds to a string(M).

Generate a list of strings and remove any strings that do not represent a TM to get a list of TMs.

Diagonalization over languages

Proof (continued)

- The set of infinite binary sequences, B, is uncountable. (Exactly the same proof we gave for uncountability of R)
- Let L be the set of all languages over Σ.
- For each language $A \in L$ there is unique infinite binary sequence X_A
  - The ith bit in X_A is 1 if $s_i \in A$, 0 otherwise.

Σ^*=	{	E,	0,	1,	00,	01,	10,	11,	000,	001,	⋯ }
A=	{		0,		00,	01,			000,	001,	⋯ }
X_A=	{	0	1	0	1	1	0	0	1	1	⋯ }

Diagonalization over languages

Proof (continued)

- The function $f : \mathsf{L} \longrightarrow \mathsf{B}$ is a correspondence. Thus L is uncountable.
- So, there are languages that can not be recognized by some TM. There are not enough TMs to go around.

The Acceptance problem is undecidable

Theorem

- $A_{TM} = \{(M, w) \mid M \text{ is a TM and } M \text{ accepts } w\}$, *is undecidable.*

- We assume A_{TM} is decidable and obtain a contradiction.
  - Suppose H decides A_{TM}

$$H((M, w)) = \begin{array}{ll} accept & \text{if } M \text{ accepts } w \\ reject & \text{if } M \text{ does not accept } w \end{array}$$

The Acceptance problem is undecidable

Proof (continued)

- We now construct a new TM D
  $D =$ "On input (M), where M is a TM
  1. Run H on input $(M, (M))$.
  2. If H accepts, D *rejects*, if H rejects, D *accepts*"
- So

$$D((M)) = \begin{array}{ll} \textit{accept} & \text{if } M \text{ does not accept } (M) \\ \textit{reject} & \text{if } M \text{ accepts } (M) \end{array}$$

- When D runs on itself we get

$$D((D)) = \begin{array}{ll} \textit{accept} & \text{if } D \text{ does not accept } (D) \\ \textit{reject} & \text{if } D \text{ accepts } (D) \end{array}$$

- Neither D nor H can exist.

Consider the behaviour of all possible deciders:

	(M_1)	(M_2)	(M_3)	(M_4)	$\cdots$	(D) (M_j)	$\cdots$
M_1	<u>accept</u>	reject	accept	reject	$\cdots$	accept	$\cdots$
M_2	accept	<u>accept</u>	accept	accept	$\cdots$	accept	$\cdots$
M_3	reject	reject	<u>reject</u>	reject	$\cdots$	reject	$\cdots$
M_4	accept	accept	reject	<u>reject</u>	$\cdots$	accept	$\cdots$
$\vdots$		$\vdots$			$\ddots$		
$D = M_j$	reject	reject	accept	accept	$\cdots$	<u>?</u>	$\cdots$
$\vdots$		$\vdots$					$\ddots$

- D computes the opposite of the diagonal entries!

A Turing unrecognizable language

A language is co-Turing-recognizable if it is the complement of a Turing-recognizable language.

A language is decidable if it is Turing-recognizable and co-Turing-recognizable.

$\overline{A_{TM}}$ is not Turing recognizable.

- We know A_{TM} is Turing-recognizable.
- If A_{TM} were also co- Turing-recognizable, A_{TM} would have to be decidable.
- We know A_{TM} is not decidable.
- A_{TM} must not be co -Turing-recognizable.